p739 timing-boundary audit

Neutral source-and-record evidence packet, 2026-10-30

Scope. Public synthetic known-answer observations only. This report records source semantics and retained ranges. It makes no performance judgment, protocol amendment, experiment approval, qualified ratio, or new solver execution.

Bound evidence

Instrument commit: `543a54b4b1c3be7052872af3ffe393a44c69a3c3`.
Record SHA-256: `9ccc07c9b55b9180af36c3b910891803dde7f4249742881a7322f748bfdb73f7`.
Protocol SHA-256: `d2ca570ce39f2effa9564576bdb6b6e16148a0a6d511eeb05c3a01565e48d948`.
The packet retains complete source blobs and their Git object IDs in `audit.json` and `source-blobs/`.

Observed interval ranges

Arm	solve wall	common method-call wall	online wall
pencil	6.413–14.484 ms; median 13.107 ms	7.479–15.533 ms; median 14.178 ms	4.678–12.692 ms; median 11.359 ms
rho	0.210–0.353 ms; median 0.265 ms	0.369–0.513 ms; median 0.423 ms	0.163–0.303 ms; median 0.216 ms
rho_AA	0.207–0.356 ms; median 0.264 ms	0.365–0.511 ms; median 0.421 ms	0.160–0.307 ms; median 0.216 ms

Each arm contains 360 rows: 300 measured and 60 warmup. All 1,080 outcomes are verified and match the public target. Pencil rows are L2. All rho and rho_AA rows are L1 and carry the solve-too-short blocker.

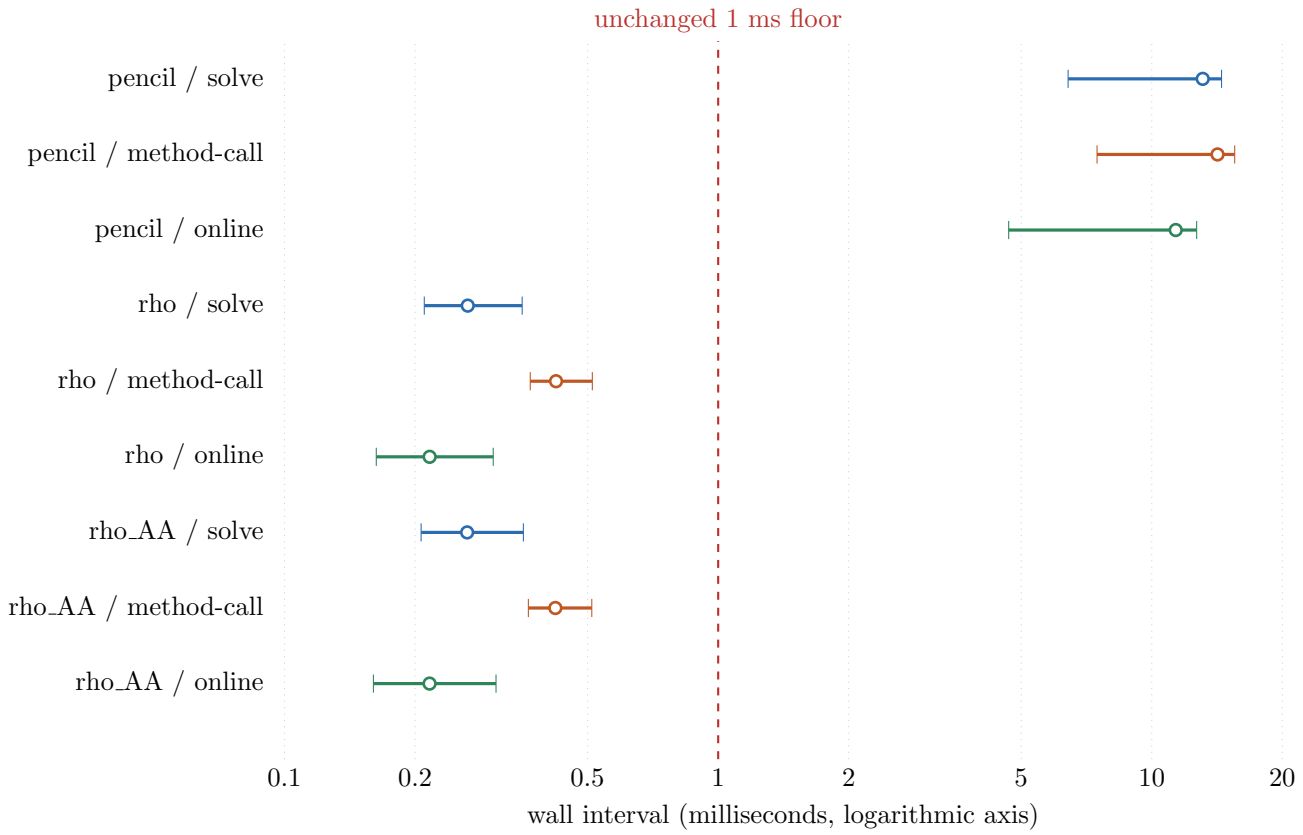


Figure. Min–max whiskers and exact median dot from all retained rows. This is descriptive only.

Observed source and protocol facts

1. Grading version 3 compares `time.solve_wall_ns` with the unchanged 1,000,000 ns floor. The protocol primary cold field is `time.method_call_wall_ns`.
2. The protocol prose says the cold endpoint is the immediate method return before the profiling dump. The selected method-call field actually stops after nested instrumentation closes and `methods::complete` returns.
3. Every one of the 720 rho/rho_AA method-call intervals is also below 1 ms. Changing only the classifier field would therefore admit none of those rows.
4. The instrument contract permits `rho_solve` and `recovery_check` for a generic method. The canonical exporter checks every arm for exactly the five index-calculus target phase names.

Qualification status. Raw intervals and operation/count evidence remain observations. Timing qualification and qualified ratios are null in this audit. Process wall is not substituted because it includes work outside the declared common method boundary.

Prospective design question; not approved

A fixed repetition count of eight was discussed because the smallest observed rho/rho_AA online interval is 160,313 ns and eight times that observation is 1,282,504 ns. This arithmetic does not approve K=8. Any repetition factor, boundary, phase schema, budget, or new run requires a new pre-data protocol and review. A follow-up would have to preserve the exact targets, arms, controls, streams and unchanged 1 ms floor; retain each repetition; establish a cold-state/reset boundary; separate one-invocation operation counts from timing amplification; and fail closed.

Limits

Ranges include warmup and measured rows, counted separately. No uncertainty interval, significance test, speed ratio, cost ratio, or ranking is computed. This is a source-and-record audit, not an independent mathematical proof or timing qualification.

Reproduction

The native Rust extractor in `native_extract/` accepts only the pinned source objects, record hash, and protocol hash. It invokes no solver. Exact commands and the packet manifest are in `README.md` and `SHA256SUMS`.